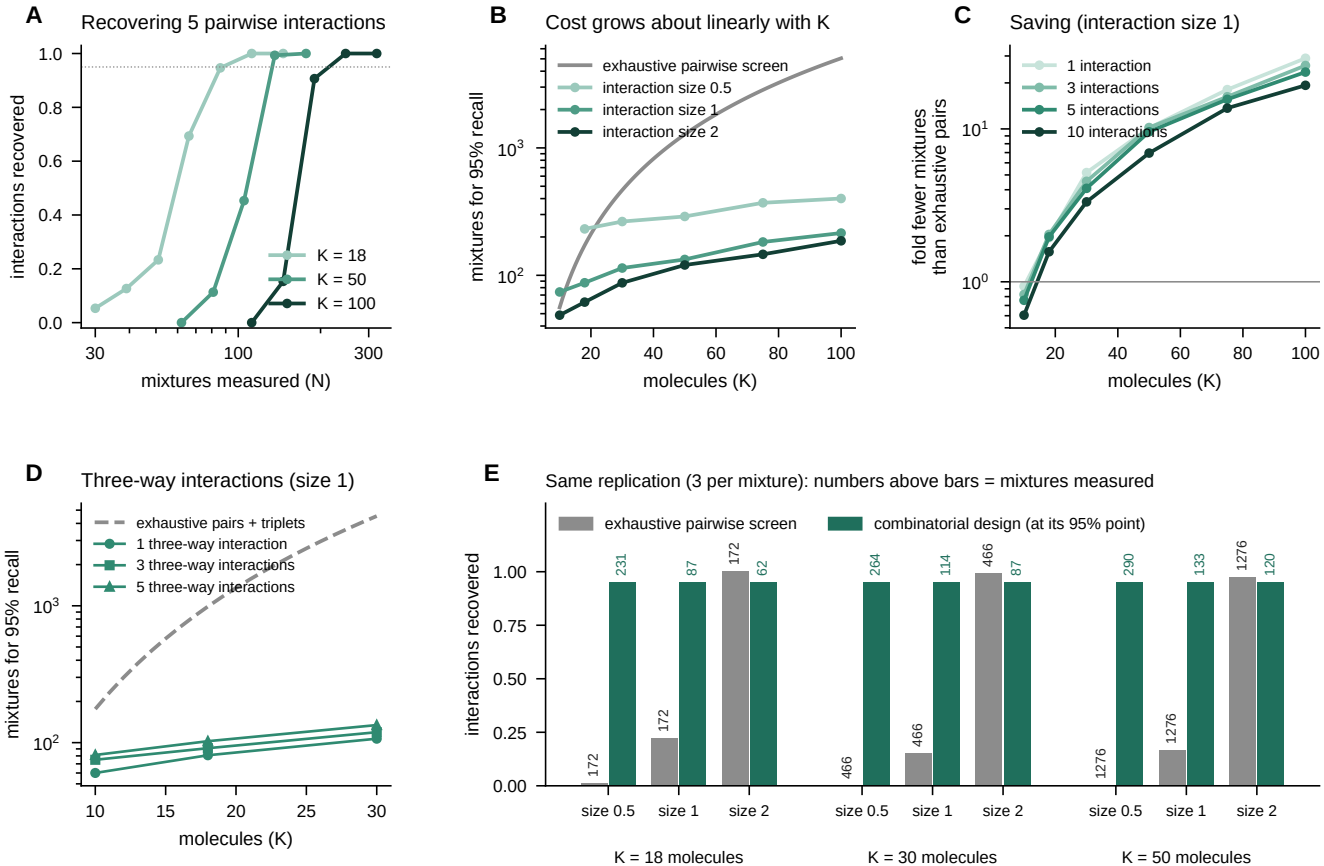
A
Recovering 5 pairwise interactions
interactions recovered
mixtures measured (N)
K = 18
K = 50
K = 100
B
Cost grows about linearly with K
mixtures for 95% recall
molecules (K)
exhaustive pairwise screen
interaction size 0.5
interaction size 1
interaction size 2
C
Saving (interaction size 1)
fold fewer mixtures than exhaustive pairs
molecules (K)
1 interaction
3 interactions
5 interactions
10 interactions
D
Three-way interactions (size 1)
mixtures for 95% recall
molecules (K)
exhaustive pairs + triplets
1 three-way interaction
3 three-way interactions
5 three-way interactions
E
Same replication (3 per mixture): numbers above bars = mixtures measured
interactions recovered
exhaustive pairwise screen
combinatorial design (at its 95% point)
size 0.5
size 1
size 2
K = 18 molecules
K = 30 molecules
K = 50 molecules